

Google Cloud Associate Cloud Engineer Syllabus

Archer Infotech, Kothrud Pune | Classroom & live-online batches | Updated September 2026

Course Overview

This syllabus lists every module of the course in teaching order, with the topics covered in each. It is generated from the course page, so the two cannot disagree.

Module 1: Exam Strategy, Project Setup & Foundations Review

Schedule: Week 1

How the GCP ACE exam scores you (Google's item-response-theory model), the question patterns Google uses (scenario-based multiple-choice and multi-select), test-day logistics for Kryterion. Foundations review — Organisation / Folder / Project hierarchy, IAM principals, the gcloud CLI essentials. By the end of week 1 every student has done a 50-question diagnostic.

Topics

- ACE exam structure and Google's grading model
- Question patterns — scenario-based MCQ / multi-select
- Test-day logistics — Kryterion (online vs centre)
- Organisation / Folder / Project hierarchy
- IAM principals and roles
- gcloud CLI essentials
- First diagnostic exam

Module 2: GCP Fundamentals Refresher — Console, gcloud & Cloud Shell

Schedule: Week 1

The Associate Cloud Engineer exam is unusually command-line heavy for an associate certification: a large share of its questions show you a gc lound invocation and ask what it does, or describe a task and ask which command performs it. Fluency in the CLI is therefore not optional, and this week builds it.

You work through the console, Cloud Shell and the SDK, configurations and named profiles, and the gc lound command grammar — group, subgroup, verb, flags — which is regular enough that you can often reconstruct a command you have never used. The resource hierarchy of organisation, folders and projects is covered here because everything in the rest of the course inherits down it.

Topics

- Regions, zones and multi-region resources
- Organisation, folders, projects and the hierarchy
- The Cloud Console and Cloud Shell
- Installing and initialising the gcloud SDK
- gcloud configurations and named profiles
- The gcloud command grammar
- gsutil, bq and the specialised tools
- Cloud APIs and enabling services
- Labels, and how they differ from tags
- Reading Google Cloud documentation efficiently

Module 3: Domain 1 — Setting Up Cloud Projects & Accounts (~17%)

Schedule: Week 2

Project setup at depth. Cover the resource hierarchy, billing setup (linked billing accounts, budget alerts), the gcloud CLI configuration, plus the patterns Google asks about — multi-project organisation, Shared VPC for centralised networking, organisation policies for governance.

Topics

- Resource hierarchy — Organisation / Folder / Project
- Billing setup and budget alerts
- gcloud CLI configuration
- Multi-project organisation
- Shared VPC
- Organisation policies

Module 4: Domain 2 — Planning & Configuring Cloud Solutions (~17%)**Schedule: Week 3**

Resource planning. Compute selection — Compute Engine (custom machine types, preemptible / spot), GKE Autopilot vs Standard, Cloud Run, Cloud Functions, App Engine. Data services — Cloud SQL vs Spanner vs Firestore vs BigQuery vs Bigtable (when each fits). Network planning — VPC design, custom vs auto subnets, regional considerations. Plus cost estimation via Pricing Calculator.

Topics

- Compute selection — GCE, GKE, Cloud Run, Functions, App Engine
- Data service selection — SQL, Spanner, Firestore, BigQuery, Bigtable
- VPC design and subnet planning
- Network costs and egress
- Pricing Calculator

Module 5: Compute Options — GCE, GKE, Cloud Run, App Engine & Functions**Schedule: Week 3**

The selection question that runs through the whole exam: given this workload, which compute product is correct. Compute Engine covers machine families, custom machine types, preemptible and spot instances, instance templates, managed instance groups and autoscaling.

GKE covers clusters, node pools, Autopilot against Standard, and the kubectl basics the exam expects. Cloud Run covers container-based serverless, App Engine covers standard and flexible environments, and Cloud Functions covers event-driven code. The module closes on the decision itself — stated as a set of criteria rather than a preference, because that is how the exam frames it.

Topics

- Machine families, custom types and right-sizing
- Preemptible and spot VMs
- Instance templates and managed instance groups
- Autoscaling and autohealing
- GKE clusters, node pools, Autopilot and Standard
- kubectl basics the exam expects
- Cloud Run and container-based serverless
- App Engine standard versus flexible
- Cloud Functions and event triggers
- Choosing a compute product from stated constraints

Module 6: Storage & Databases — Cloud Storage, SQL, Firestore & BigQuery

Schedule: Week 4

The data half of the exam, and another selection problem. Cloud Storage covers buckets, storage classes and their retrieval costs, lifecycle rules, object versioning, retention policies and signed URLs — plus uniform access control, which the exam asks about directly.

Databases cover Cloud SQL with high availability and read replicas, Cloud Spanner and what its horizontal scale is for, Firestore for document data, Bigtable for wide-column time-series work, Memorystore for caching, and BigQuery for analytics. Persistent disks and Filestore complete the block and file picture. The recurring exam shape is a workload description that maps to exactly one of these, and the module trains recognising it.

Topics

- Cloud Storage buckets and storage classes
- Lifecycle rules, versioning and retention
- Uniform versus fine-grained access control
- Signed URLs and signed policy documents
- Persistent disks, local SSD and Filestore
- Cloud SQL, high availability and read replicas
- Cloud Spanner and horizontal relational scale
- Firestore and Bigtable — different shapes of data
- Memorystore for caching
- BigQuery for analytics, and its pricing model

Module 7: Networking — VPC, Firewall Rules, Load Balancing & DNS**Schedule: Week 4**

Google Cloud's networking model differs from its competitors in ways the exam tests deliberately: VPCs are global, subnets are regional, and firewall rules are stateful with priorities and target tags. Candidates carrying assumptions from AWS or Azure lose marks here, so the differences are made explicit.

You build VPCs and subnets, write firewall rules with priorities and service accounts as targets, and configure Cloud NAT for private instances. Load balancing covers the global and regional options and the layer each operates at, then Cloud DNS, Cloud CDN, VPC peering, Shared VPC and the hybrid connectivity options at the level the exam expects.

Topics

- Global VPCs and regional subnets
- Auto mode versus custom mode networks
- Firewall rules, priorities, tags and service accounts
- Cloud NAT and private instances
- Private Google Access
- Global and regional load balancing options
- Cloud DNS and private zones
- Cloud CDN and caching at the edge
- VPC peering and Shared VPC
- Cloud VPN and Interconnect at exam level

Module 8: Domain 3 — Deploying & Implementing Solutions (~25%)**Schedule: Weeks 4–5**

The largest exam domain. Compute deployment — Compute Engine instances, instance templates, Managed Instance Groups + autoscaling, GKE deployment (Autopilot + Standard), Cloud Run deployment, Cloud Functions deployment, plus container image management via Artifact Registry. Data deployment — Cloud SQL setup, BigQuery dataset / table creation, Cloud Storage buckets and lifecycle. Plus deployment via Deployment Manager (the GCP-native IaC) and Terraform.

Topics

- Compute Engine — instances, templates, MIGs
- GKE Autopilot and Standard deployment
- Cloud Run deployment
- Cloud Functions deployment
- Artifact Registry
- Cloud SQL setup
- BigQuery dataset / table
- Cloud Storage and lifecycle
- Deployment Manager and Terraform

Module 9: Domain 4 — Ensuring Successful Operation (~20%)

Schedule: Week 6

Operations domain. Monitoring — Cloud Monitoring metrics, alert policies, dashboards. Logging — Cloud Logging, log-based metrics, log routing (sinks). Cloud Trace and Cloud Profiler. Plus the maintenance patterns — scaling, backups (Cloud SQL automated backups, Cloud Storage versioning), troubleshooting flowcharts.

Topics

- Cloud Monitoring metrics and alert policies
- Cloud Logging and log-based metrics
- Log sinks for routing
- Cloud Trace and Profiler
- Backup patterns
- Troubleshooting flowcharts

Module 10: Operations — Cloud Monitoring, Logging & Error Reporting

Schedule: Week 6

Domain 4 rests on the operations suite, and the exam asks practical questions: which tool shows you this, and how would you alert on it. Cloud Monitoring covers metrics, dashboards, uptime checks, alerting policies and notification channels, plus the Ops Agent that has to be installed before VM memory and disk metrics exist at all — a detail that appears in questions.

Cloud Logging covers the log explorer, query syntax, log sinks and exports to Cloud Storage or BigQuery, retention and log-based metrics. Error Reporting, Cloud Trace and Cloud Profiler close the module, along with the diagnostic habit the exam rewards: identifying which signal answers the question before reaching for a tool.

Topics

- Cloud Monitoring metrics and dashboards
- Uptime checks and alerting policies
- Notification channels and escalation
- The Ops Agent and VM-level metrics
- Cloud Logging and the log explorer
- Log query syntax and filters
- Log sinks, exports and retention
- Log-based metrics
- Error Reporting, Cloud Trace and Profiler
- Choosing the signal before choosing the tool

Module 11: Billing, Quotas & Cost Management

Schedule: Week 6

A small domain by weight and an easy one to secure, provided you have actually looked at a billing console rather than read about one. Billing accounts and their relationship to projects, budgets and alerts, billing export to BigQuery, and the reports

that answer where the money went.

Quotas get their own treatment because they are a frequent exam subject and a frequent real obstacle: the difference between rate and allocation quotas, how to see current usage, and how an increase is requested. Committed use and sustained use discounts close the module, along with preemptible instances as a cost strategy and the pricing calculator.

Topics

- Billing accounts and project association
- Budgets, thresholds and alerting
- Billing export to BigQuery
- Cost breakdown and reports
- Rate quotas versus allocation quotas
- Viewing usage and requesting increases
- Committed use discounts
- Sustained use discounts
- Preemptible instances as a cost strategy
- The Google Cloud pricing calculator

Module 12: Domain 5 — Configuring Access & Security (~21%)

Schedule: Week 7

Access and security. IAM — predefined vs custom roles, conditions, service accounts (the GCP differentiator from AWS / Azure — service accounts are first-class principals), Workload Identity Federation. VPC Service Controls. Plus encryption — Google-managed keys, customer-managed encryption keys (CMEK), customer-supplied encryption keys (CSEK).

Topics

- IAM predefined vs custom roles
- IAM conditions
- Service accounts as first-class principals
- Workload Identity Federation
- VPC Service Controls
- Encryption — Google-managed, CMEK, CSEK

Module 13: IAM in Depth — Roles, Service Accounts & Org Policy

Schedule: Week 7

The largest identity component of the exam, and the one with the most Google-specific behaviour. The three role types — basic, predefined and custom — with the standing guidance that basic roles are almost never the correct exam answer. Policy inheritance down the resource hierarchy, and the union behaviour that means a permission granted higher up cannot be removed lower down by an allow policy.

Service accounts get extended treatment: default against user-managed, impersonation, key management and the strong preference for avoiding downloaded keys, and Workload Identity for GKE. Organisation policy constraints, IAM Conditions and the audit questions the exam asks close the module.

Topics

- Basic, predefined and custom roles
- Why basic roles are rarely the right answer
- Policy inheritance and the union rule
- Members, bindings and allow policies
- Service accounts — default and user-managed
- Service account impersonation
- Key management, and avoiding downloaded keys
- Workload Identity for GKE

- Organisation policy constraints
- IAM Conditions and audit logging

Module 14: Mock Exams & Test-Day Prep

Schedule: Week 8

Two full-length mock exams under timed conditions — students who score 75–85% on the second mock typically pass first attempt. Final gap-closure on weak domains, exam-booking guidance for Kryterion (online vs test centre), test-day strategy.

Topics

- Mock exam 1 — full timed
- Detailed answer review
- Mock exam 2 — full timed
- Final gap-closure
- Kryterion booking — online vs test centre
- Test-day strategy

Module 15: AI-Assisted Development Workflow

Schedule: Final Week

The skill every 2026 hiring panel now probes for — building real work with AI in the loop, responsibly. Learn to drive AI assistants (GitHub Copilot, Claude, Cursor, and IDE-native AI) to scaffold and accelerate the tools and stack this course covers, generate tests, explain and refactor unfamiliar code, and cut the boilerplate — while keeping you firmly in control of every decision. Heavy focus on guardrails: reviewing each AI suggestion, spotting hallucinated APIs or wrong answers, and handling licensing and data-privacy concerns. Close with a mini-project that takes a deliverable end-to-end using an AI-assisted workflow, then fold the same tooling into version control and everyday team practice.

Topics

- AI assistants — GitHub Copilot, Claude, Cursor, IDE-native AI
- Effective prompting for this course's stack — scaffolding, boilerplate, config
- AI-assisted test generation and coverage
- Explaining, refactoring, and modernising unfamiliar code with AI
- AI-driven review, error detection, and quality checks
- Generating and maintaining documentation with AI
- AI debugging — interpreting errors, logs, and failing output
- Guardrails — reviewing output, avoiding hallucinations, licensing & data privacy
- Team workflow — AI in the editor, in reviews, and in delivery pipelines
- Mini-project — a deliverable built end-to-end with an AI-assisted workflow

Enquire or book a free demo session. Archer Infotech, Flat No. 12, Divyadarshan Housing Society, behind Kothrud Bus Stand Road, Kothrud, Pune 411038. Phone +91 9850 678451. Email info@archerinfotech.in. Web archerinfotech.in.